

Performance Testing

Date	09 July 2026
Team ID	SWTID-2026-5556
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing, Chrome Browser
Type of Testing	Functional Testing, Performance Testing
Target Module / API	Flask Web Application (Credit Card Approval Prediction)
Test Environment	Localhost (Windows 11, Python 3.x, Flask)
Test Date	09 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Open Home Page	1	10 sec	Page loads successfully
2	Enter Applicant Details	1	20 sec	Input accepted successfully
3	Click Predict Button	1	5 sec	Prediction generated

4	Display Result	1	5 sec	Approved/Rejected displayed
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.3 sec	Pass	Acceptable
3	Throughput (Req/sec)		Good	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	35%	Pass	Efficient
6	Memory Utilization	< 80%	42%	Pass	Normal

Step 4: Observations & Analysis

Key Findings

The Flask application processed applicant data quickly and generated credit card approval predictions with low response time and no functional errors.

Bottlenecks Identified

No major bottlenecks were observed during testing.

Optimization Steps Taken

Optimized the machine learning model loading, improved preprocessing efficiency, and validated user inputs to reduce processing time.

Step 5: Screenshots / Evidence



Credit Card Approved

Congratulations! Based on our predictive model evaluation, the profile exhibits credit indicators aligning with standard risk thresholds.

Evaluation Metrics

Credit Class	System Confidence
Low Risk Profile	95.6% (XGBoost)
Debt Ratio	Recommendation
Safe Bracket	Issue Credit Limit

Suggested Next Steps

- Select preferred interest tiers and card programs.
- Verify identity credentials with nearest branch office.
- Complete security enrollment on mobile app.

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